CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

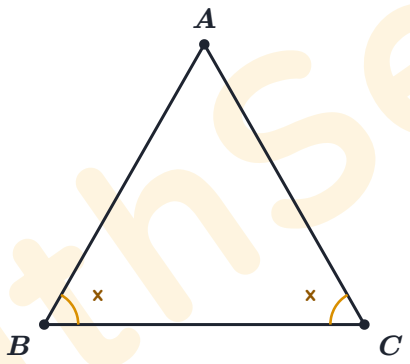
Chapter 10: Isosceles Triangles

Topic-wise short notes with solved examples

The two basic theorems on isosceles triangles and their converse, the properties of equilateral triangles, and angle-chasing problems that use them.

1 The isosceles triangle

- A triangle with **two equal sides** is **isosceles**. The third side is the **base**, the angles on the base are the **base angles**, and the angle between the equal sides is the **vertical angle**.
- **Theorem 1**: if two sides of a triangle are equal, the **angles opposite to them are equal**.
- **Theorem 2 (converse)**: if two angles of a triangle are equal, the **sides opposite to them are equal**.



AB = AC, so the base angles B and C are equal

- **Proof of Theorem 1**: draw AD, the bisector of $\angle A$, meeting BC at D. In $\triangle ABD$ and $\triangle ACD$: $AB = AC$, $\angle BAD = \angle CAD$, $AD = AD$. So they are congruent (S.A.S.) and $\angle B = \angle C$.
- If the vertical angle is v, each base angle is $\frac{180^\circ - v}{2}$. If each base angle is b, the vertical angle is $180^\circ - 2b$.

Example 1.1 In triangle ABC, $AB = AC$ and $\angle A = 70^\circ$. Find $\angle B$ and $\angle C$.

SOLUTION

$$\angle B = \angle C = \frac{180^\circ - 70^\circ}{2}$$

55° **each**

Example 1.2 In triangle PQR, $\angle P = 40^\circ$ and $\angle Q = 100^\circ$. Which sides are equal?

SOLUTION

$\angle R = 180^\circ - 140^\circ = 40^\circ = \angle P$. Sides opposite equal angles are equal.

$QR = PQ$

2 Properties of an isosceles triangle

- In triangle ABC with $AB = AC$, the following four lines from A are **the same line**: the bisector of $\angle A$, the median to BC, the altitude to BC, and the perpendicular bisector of BC.
- So the altitude from the vertex **bisects the base** and **bisects the vertical angle**.
- The medians to the equal sides are equal; so are the altitudes to the equal sides and the bisectors of the base angles.

Example 2.1 In triangle ABC, $AB = AC = 13$ cm and $BC = 10$ cm. Find the altitude from A.

SOLUTION

The altitude AD bisects BC, so $BD = 5$ cm. $AD^2 = 13^2 - 5^2 = 144$.

12 cm

Example 2.2 Prove that the altitudes to the equal sides of an isosceles triangle are equal.

SOLUTION

Let $AB = AC$, with BE perpendicular to AC and CF perpendicular to AB. In $\triangle BEC$ and $\triangle CFB$: $\angle BEC = \angle CFB = 90^\circ$, $\angle ECB = \angle FBC$ (base angles), $BC = CB$.

The triangles are congruent (A.A.S.), so $BE = CF$

3 Equilateral triangles

- A triangle with all three sides equal is **equilateral**. Each of its angles is 60°.
- Conversely, a triangle with all angles equal (each 60°) is equilateral.
- An isosceles triangle with one angle of 60° is equilateral.
- An **isosceles right triangle** has angles $90^\circ, 45^\circ, 45^\circ$.

Example 3.1 In an isosceles triangle the vertical angle is twice a base angle. Find the angles.

SOLUTION

Let each base angle be x. Then $x + x + 2x = 180^\circ$.

$45^\circ, 45^\circ, 90^\circ$

4 Angle-chasing

- Mark equal sides, then mark the equal angles opposite them.
- Use the angle sum of a triangle, the exterior angle property and angles on a straight line.
- Give each unknown angle a letter and form an equation.

Example 4.1 In triangle ABC, $AB = AC$. Side BC is produced to D and $\angle ACD = 115°$. Find the angles of the triangle.

SOLUTION

$\angle ACB = 180° - 115° = 65° = \angle ABC$

$\angle A = 180° - 130°$

$65°, 65°, 50°$

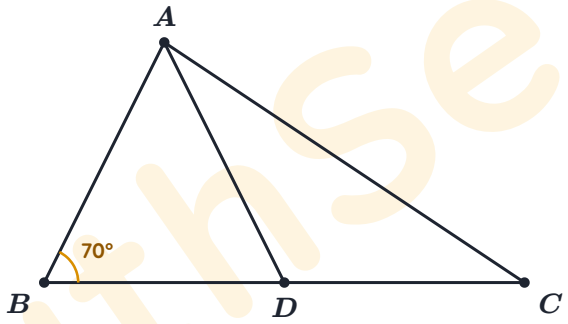
Example 4.2 In the figure, D is a point on BC such that $AB = AD = DC$ and $\angle B = 70°$. Find $\angle C$.

SOLUTION

$AB = AD$ gives $\angle ADB = \angle B = 70°$.

$\angle ADB$ is an exterior angle of $\triangle ADC$, and $AD = DC$ gives $\angle DAC = \angle C$. So $2\angle C = 70°$.

$\angle C = 35°$



AB = AD = DC

Example 4.3 ABC is an equilateral triangle and BCDE is a square drawn outside it. Find $\angle BAE$.

SOLUTION

$\angle ABE = 60° + 90° = 150°$, and $AB = BC = BE$, so triangle ABE is isosceles.

$$\angle BAE = \frac{180° - 150°}{2}$$

$15°$

Example 4.4 In triangle ABC, $AB = AC$ and the bisectors of $\angle B$ and $\angle C$ meet at O. Prove that $OB = OC$.

SOLUTION

$\angle B = \angle C$, so $\frac{1}{2}\angle B = \frac{1}{2}\angle C$, that is $\angle OBC = \angle OCB$.

Sides opposite equal angles are equal, so $OB = OC$

5 Common mistakes

- Marking the wrong angles as equal. The equal angles are **opposite** the equal sides.
- Assuming a triangle is isosceles because it looks so in the figure.
- Using the bisector, median and altitude as the same line in a triangle that is not isosceles.
- Forgetting that the result works both ways: equal sides give equal angles, and equal angles give equal sides.
- Taking the given angle as a base angle when it is the vertical angle, or the other way round.

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